MS1Vtx_rms_clustime_raw

MC vs data · overview · rank 40 / 98


```
rank 40 / 98 by endcap D_equal

used by: NN2 barrel, NN2 endcap

rank-identical columns merged into this one:
    MS1Vtx_barrel_rms_clustime
    MS1Vtx_endcap_rms_clustime
```

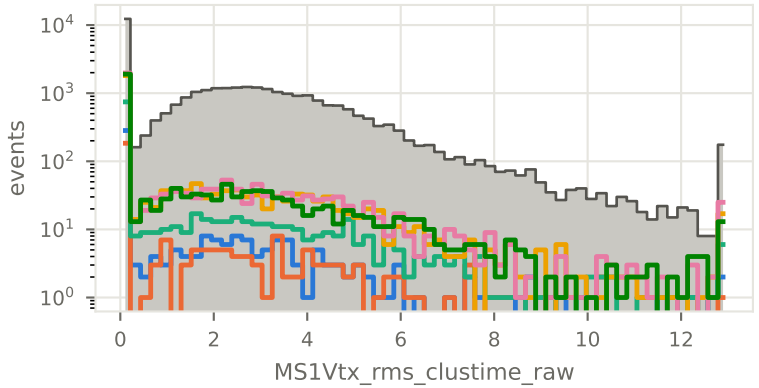
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.335	0.022	0.341	0.342	0.309	0.348	0.338	0.335	0.356	34,970	40/98
barrel	0.262	0.051	0.268	0.317	0.218	0.261	0.282	0.284	0.249	1,089	50/98
all	0.337	0.018	0.344	0.355	0.292	0.346	0.346	0.347	0.347	36,059	40/98

MS1Vtx_rms_clustime_raw

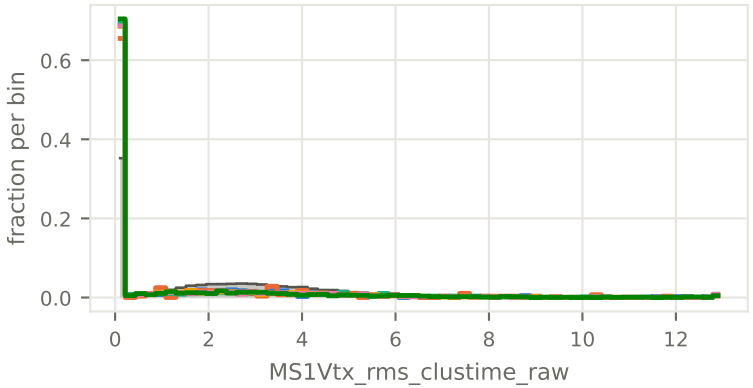
region: ENDCAP · rank 40 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.335 (noise 0.022) D_pooled 0.341 min 0.309 (mH125_mS55) max 0.356 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

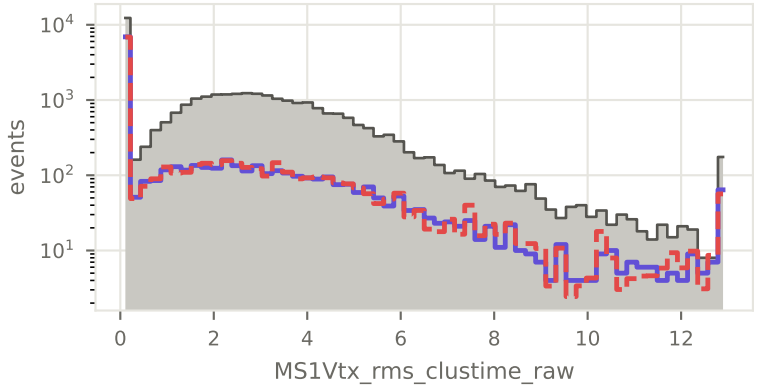


1b · each mass point vs background · normalized

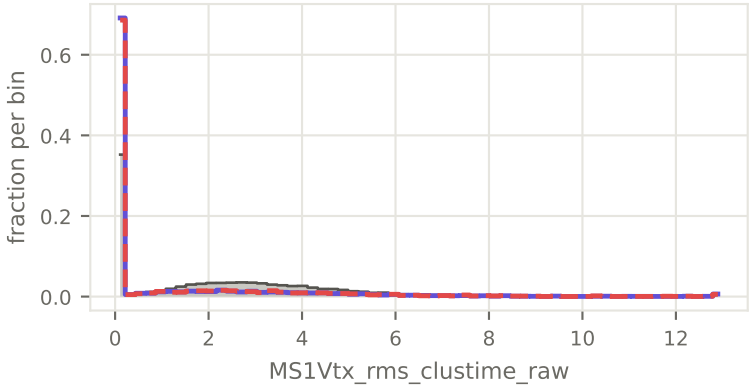


1 · each mass point
background n 34,970
mH125_mS15 D 0.342 n 411
mH125_mS55 D 0.309 n 281
mH200_mS50 D 0.348 n 1,071
mH400_mS100 D 0.338 n 2,648
mH600_mS150 D 0.335 n 2,882
mH1000_mS275 D 0.356 n 2,703

2a · combined signal vs background · raw

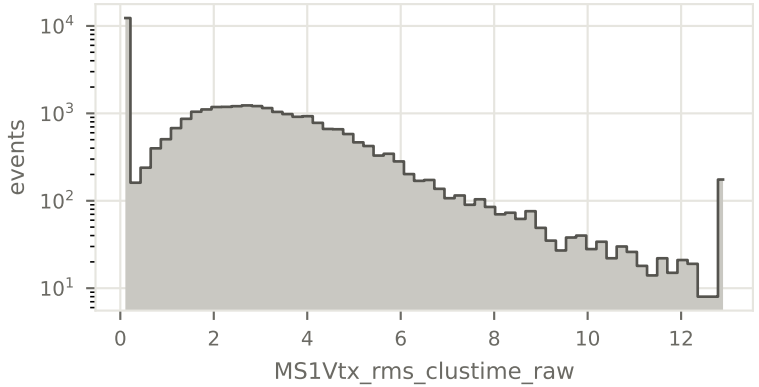


2b · combined signal vs background · normalized

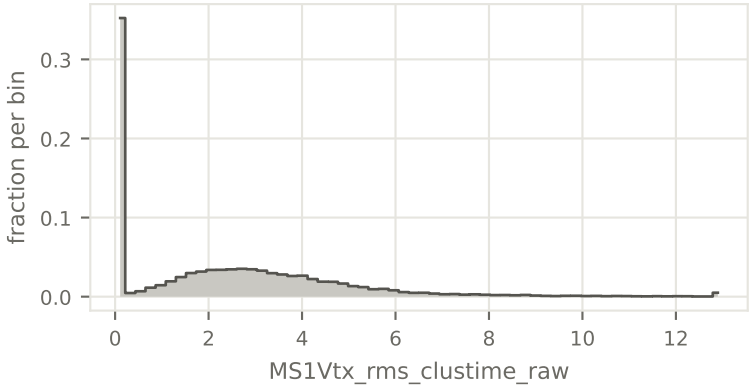


2 · combined signal
background n 34,970
combined, pooled D 0.341 n 9,996
equal weight per mass D 0.335

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

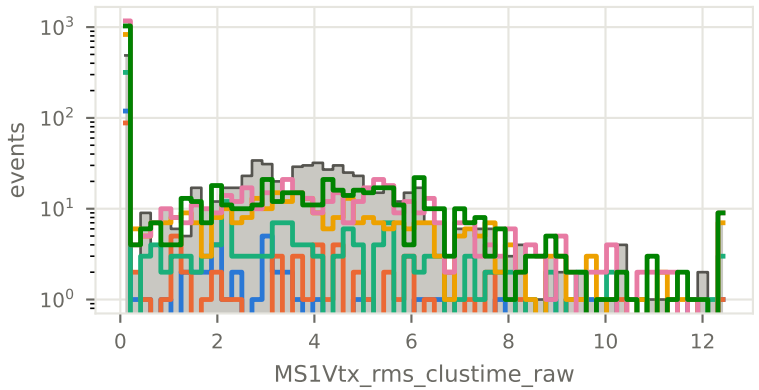
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_rms_clustime_raw

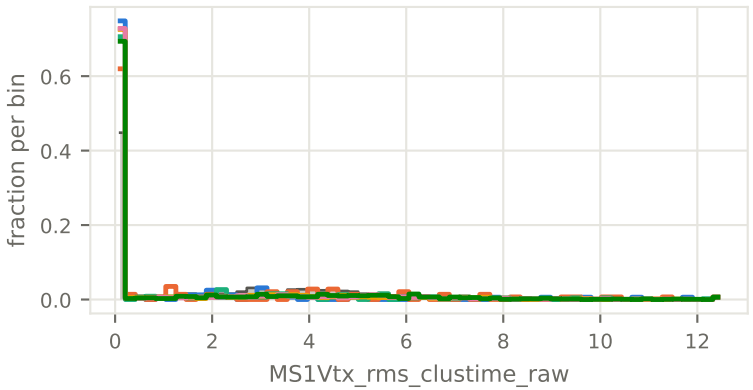
region: BARREL · rank 40 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.262 (noise 0.051) D_pooled 0.268 min 0.218 (mH125_mS55) max 0.317 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

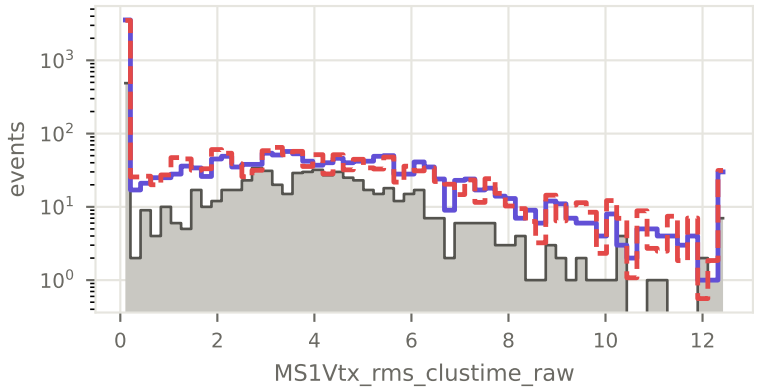


1b · each mass point vs background · normalized

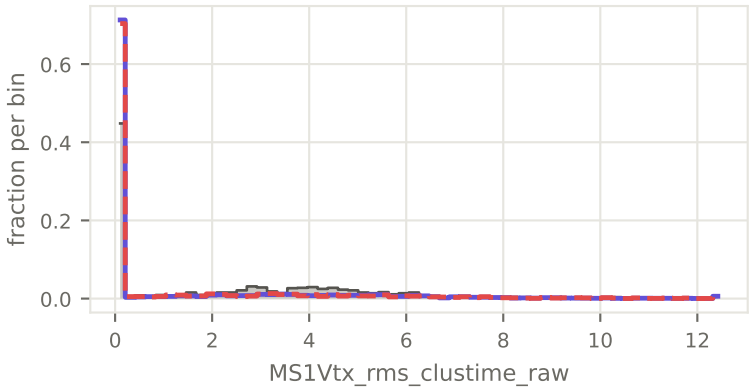


1 · each mass point
- background n 1,089
- mH125_mS15 D 0.317 n 159
- mH125_mS55 D 0.218 n 142
- mH200_mS50 D 0.261 n 449
- mH400_mS100 D 0.282 n 1,144
- mH600_mS150 D 0.284 n 1,608
- mH1000_mS275 D 0.249 n 1,485

2a · combined signal vs background · raw

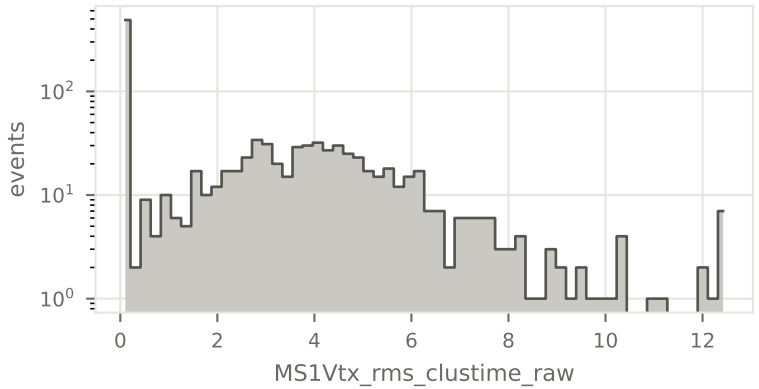


2b · combined signal vs background · normalized

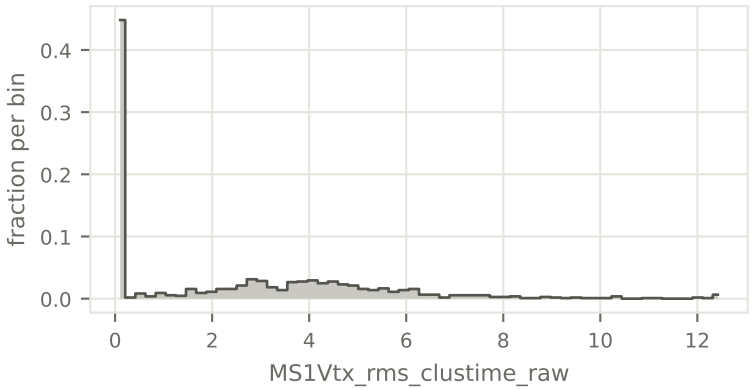


2 · combined signal
- background n 1,089
- combined, pooled D 0.268 n 4,987
- equal weight per mass D 0.262

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
- background n 1,089

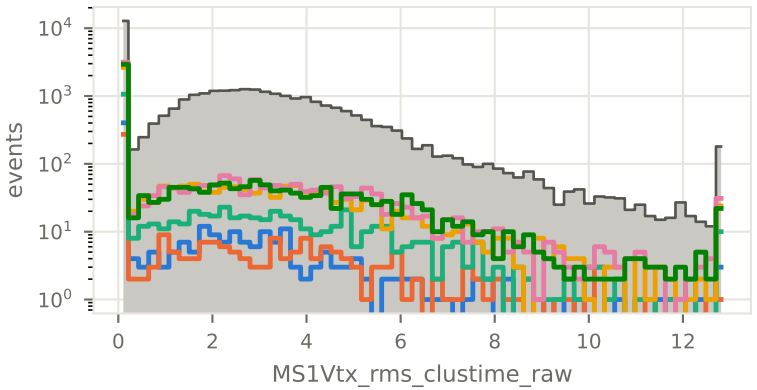
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_rms_clustime_raw

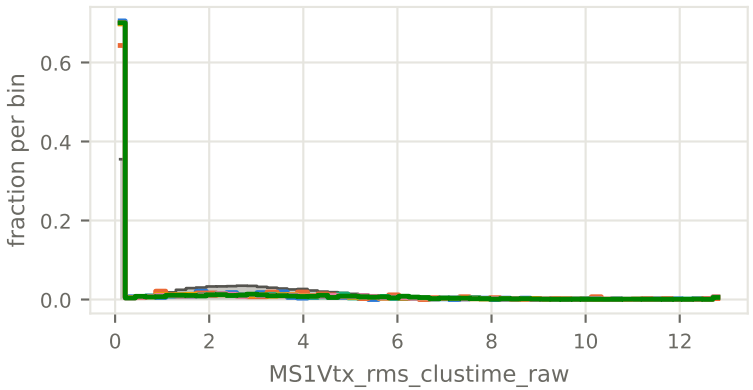
region: ALL · rank 40 / 98 · in: NN2 barrel, NN2 endcap

D_equal 0.337 (noise 0.018) D_pooled 0.344 min 0.292 (mH125_mS55) max 0.355 (mH125_mS15) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

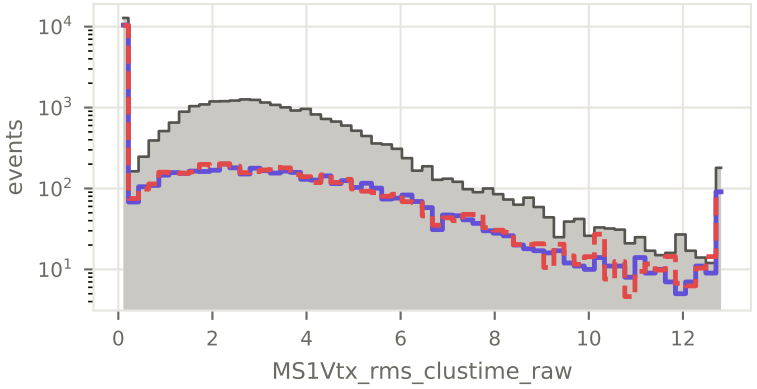


1b · each mass point vs background · normalized

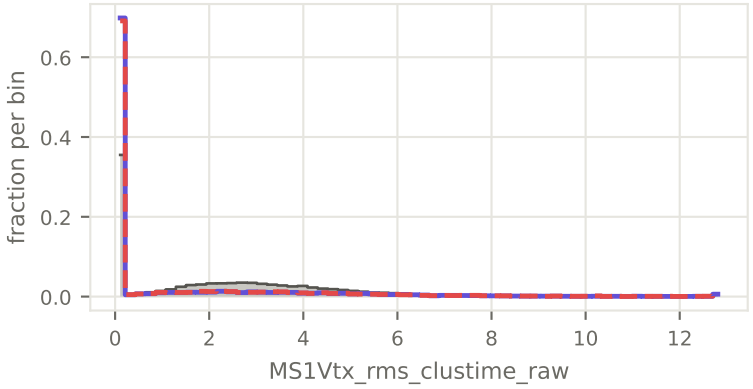


1 · each mass point
- background n 36,059
- mH125_mS15 D 0.355 n 570
- mH125_mS55 D 0.292 n 423
- mH200_mS50 D 0.346 n 1,520
- mH400_mS100 D 0.346 n 3,792
- mH600_mS150 D 0.347 n 4,490
- mH1000_mS275 D 0.347 n 4,188

2a · combined signal vs background · raw

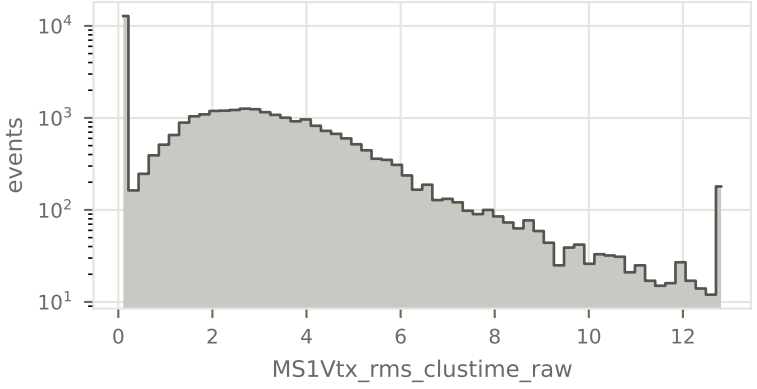


2b · combined signal vs background · normalized

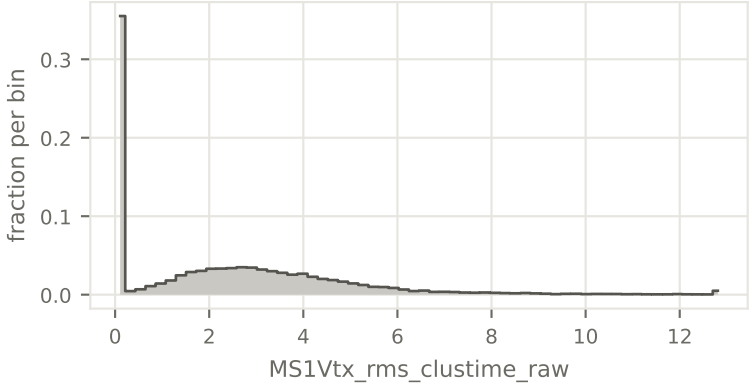


2 · combined signal
- background n 36,059
- combined, pooled D 0.344 n 14,983
- equal weight per mass D 0.337

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
- background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed